

Foundational Research & Advances in B92 Quantum Key Distribution Protocol

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1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for B92 Quantum Key Distribution Protocol in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Instead of using 4 different laser settings like BB84, B92 achieves 100% spy-proof security using only TWO non-orthogonal flashlight beams ($|0\rangle$ and $|+\rangle$)!"

3. Mathematical Formulation & Unitary Rule

$$\langle 0 | + \rangle = \frac{1}{\sqrt{2}}, \quad P(\text{conclusive}) = \frac{1}{4}, \quad P(\text{inconclusive}) = \frac{3}{4}$$

Non-orthogonality prevents exact cloning; conclusive outcomes provide unambiguous bit values.

4. Step-by-Step Circuit Sequence

Alice sends $|0\rangle$ or $|+\rangle$; Bob measures in Z or X basis; public announcement of conclusive detection slots.

5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit

# Alice prepares bit 1 (|+> state)
qc = QuantumCircuit(1, 1)
qc.h(0) # Prepare |+> state
# Bob measures in Z-basis to detect conclusive |1>
qc.measure(0, 0)
print("B92 conclusive measurement simulated.")
```

6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

Vulnerable to quantum eavesdropping (information theoretic security with Foundational sifting efficiency)

7. Key Applications & Future Research Directions

- Low-complexity satellite and fiber quantum communication links
- Pedagogical demonstration of non-orthogonal quantum state security
- Quantum random beacon protocols

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant Cryptography pipelines.